Perfect gas process

Remember

1] $PV = mRT$ ⟹ T in Kelvin

2] $Pv = RT$ ⟹ sp. form < $v \to \frac{m^3}{kg}$, $R \to \frac{J}{kgK}$

3] $PV = nR_oT$ ⟹ Molar form

4] $R = \frac{R_o}{M}$; R_o → universal gas constant, M → Molar Mass

5] $m = n * M$
- m → mass in kg
- n → # of moles
- M → Molar mass

6] # of molecules $= n * A_N$
- A_N → Avogadros Number (constant) $6.023 * 10^{23}$

7] $Q = M C_v \Delta T$ → under constant volume
$= M C_p \Delta T$ → " " pressure

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$$C_p > C_v$$

8 $C_p - C_v = R$

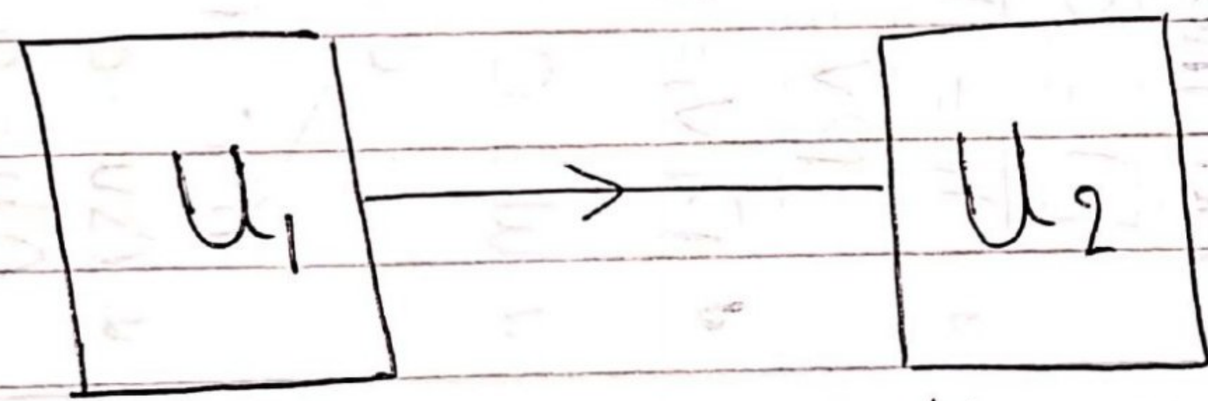
9 $\gamma = \frac{C_p}{C_v}$

10 $C_v = \frac{R}{\gamma - 1}$

11 $C_p = \gamma C_v = \frac{\gamma R}{\gamma - 1}$

For perfect gas

internal energy $\boxed{U = m C_v T}$

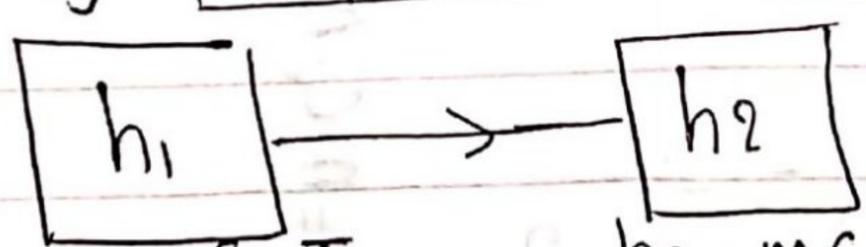


$U_1 = m C_v T_1$ $\qquad$ $U_2 = m C_v T_2$

$$\Delta U = U_2 - U_1 = m C_v (T_2 - T_1)$$

$$\boxed{\Delta U = m C_v \Delta T}$$

enthalpy $\boxed{h = pv + u}$ $\quad$ $\boxed{h = m C_p \Delta T}$



$h_1 = m C_p T_1$ $\qquad$ $h_2 = m C_p T_2$

$$\boxed{\Delta h = m C_p \Delta T}$$

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$\Delta h = m C_p \Delta T$

perfect gas process

$PV = mRT$

$Q = W + \Delta u$

Const-Volume Rigid-Cylinder

$V = C$

* $V_1 = V_2$
* $\frac{P_1}{P_2} = \frac{T_1}{T_2}$
* $W = 0$
* $Q = \Delta u = m C_v \Delta T$

Const-pressure isobaric

$P = C$

* $P_1 = P_2$
* $\frac{V_1}{V_2} = \frac{T_1}{T_2}$
* $W = P(V_2 - V_1)$
  $= (h_2 - h_1) - \Delta u$
  $= Q - \Delta u$
* $Q = m C_p \Delta T$

Const-T iso Thermal

$\boxed{T = C}$

* $T_1 = T_2$

$\boxed{PV = C}$

* $P_1 V_1 = P_2 V_2$
* $W = C \ln \frac{V_2}{V_1}$
  $= C \ln \frac{P_1}{P_2}$

  $P_1 V_1 \cdot P_2 V_2$
* $\Delta u = 0$
  $\Delta h = 0$
* $\boxed{Q = W}$

$Q = 0$ adiapatic isentropic Thermally isolated

* $Q = 0$
* $\boxed{PV^{\gamma} = C}$
* $\frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^{\gamma}$
* $\frac{T_1}{T_2} = \left(\frac{P_2}{P_1}\right)^{\frac{1-\gamma}{\gamma}}$
* $W = -\Delta u$
  $= m C_v (T_1 - T_2)$
  $= \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$

Polytropic

$\boxed{PV^{n} = C}$

$n > 1$

* $P_1 V_1^{n} = P_2 V_2^{n}$
* $W = \frac{P_1 V_1 - P_2 V_2}{n - 1}$
  $= \frac{mR(T_1 - T_2)}{n - 1}$
* $Q = W\left(\frac{\gamma - n}{\gamma - 1}\right)$

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To solve problems

1. get C_v, C_p, γ, R
2. get m
3. get P, V, T for each state
4. get W, Q for each process
5. draw P-V & T-S diagram.

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II one kg of oxygen (molar mass 32 kg/kmole, $\gamma = 1.4$) at a pressure of 25 bar is cooled at constant volume until the pressure falls to 15 bar. it is then compressed isothermally and finally, it is heated at constant pressure to its initial state. if the initial specific volume is 0.06 m^3/kg. Calculate the net work done, the net heat flow, and the overall change of entropy. Sketch the cycle on both P-V and T-S diagram.

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$m = 1\,kg$, $M = 32\ kg/kmole$

$\gamma = 1.4$

$P_1 = 25 * 10^2\ kPa$
$v_1 = 0.06\ \frac{m^3}{kg}$
$V_1 = 0.06\ m^3$
$T_1 = 577.36\ K$

$V = C$
$V_1 = V_2$
$W = 0$
$Q_1 = m C_v (T_2 - T_1)$

$P_2 = 15 * 10^2\ kPa$
$V_2 = 0.06\ m^3$
$T_2 = 346.42\ °K$

$P = C$

$P_1 = P_3$

$W_3 = P(V_1 - V_3)$

$Q_3 = m C_p (T_1 - T_3)$

$P_3 = 25 * 10^2\ kPa$
$V_3 = 0.036\ m^3$
$T_3 = 346.49\ K$

isothermal
$T_3 = T_2$
$Q_3 = W_2 = P V \ln \frac{V_3}{V_2}$

$$R = \frac{R_o}{M} = \frac{8.314}{32} = 0.2598\ \frac{kJ}{kg\,K}$$

$$C_v = \frac{R}{\gamma - 1} = \frac{0.2598}{1.4 - 1} = 0.6495\ \frac{kJ}{kg\,K}$$

$$C_p = \gamma C_v = 1.4 * 0.6495 = 0.9\ \frac{kJ}{kg\,K}$$

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State 1 $\Longrightarrow$ $P_1 V_1 = m R T_1$

$P_1 \to 25 \times 10^2$, $V_1 \to 0.06$, $m \to 1$, $R \to 0.2598$

$$T_1 = 577.367\,°K$$

State 2 $\Longrightarrow$ $P_2 V_2 = m R T_2$

$P_2 \to 5 \times 10^2$, $V_2 \to 0.06$, $m \to 1$, $R \to 0.2598$

$$T_2 = 346.42\,°K$$

$T_3 \to 346.4\,°K$

State 3 $\Longrightarrow$ $P_3 V_3 = m R T_3$

$P_3 \to 25 \times 10^2$, $V_3 \to 0.036$, $m \to 1$, $R \to 0.2598$

$$V_3 = 0.036\,m^3$$

$W_1 = 0$

$W_2 = P_1 V_1 \ln \frac{V_3}{V_2} = -45\,KJ$

$W_3 = P_1 (V_1 - V_3) = 60\,KJ$

$$W_{Net} = 15\,KJ$$

$Q_1 = m C_v (T_2 - T_1) = -150\,KJ$

$Q_2 = -45.9\,KJ$

$Q_3 = m C_p (T_1 - T_3) = 207.8\,KJ$

$$Q_{Net} = 12\,KJ$$

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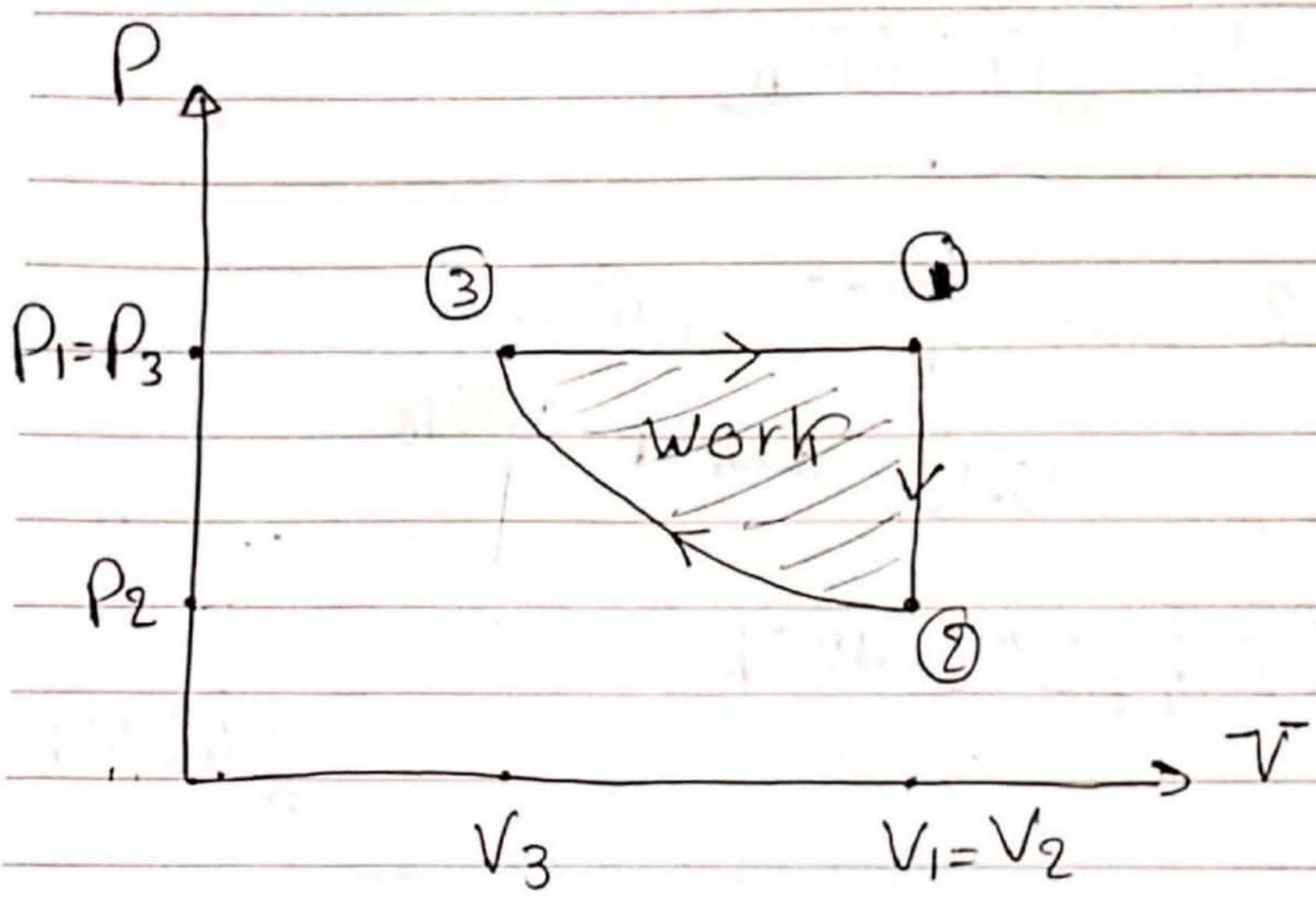
$$\boxed{Q_{net} \simeq W_{net}}$$



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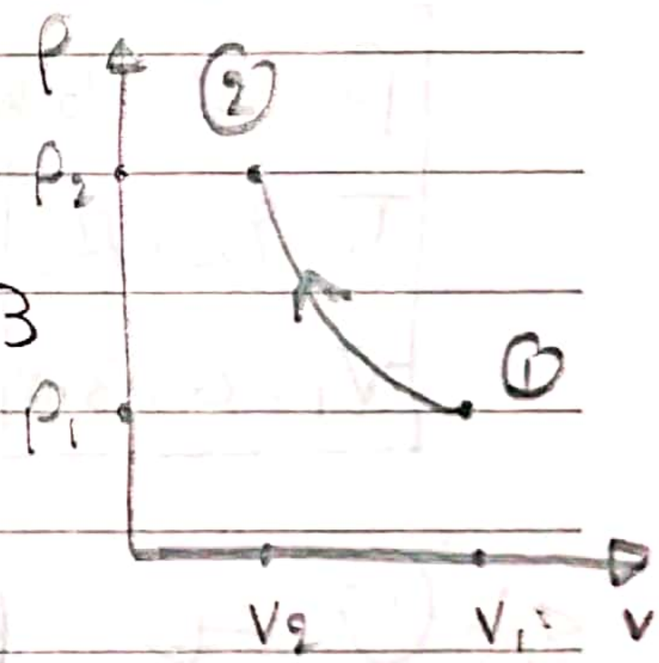
12 $m = 1\,kg$, air

- $\gamma = 1.4$
- $C_p = 1.005\ kJ/kg\,K$
- $C_v = 0.718\ kJ/kg\,K$
- $R = 0.287\ kJ/kg\,K$

St ①		St ②
$P_1 = 1.02 \times 10^2\ kPa$ $T_1 = 20 + 273 = 293\ K$	$\xrightarrow{\text{polytropic}}$ $PV^{1.3} = C$	$P_2 = 5.5 \times 10^2\ kPa$



St ① $\Rightarrow$ $P_1 V_1 = m R T_1$

$P_1 \to 1.02 \times 10^2$, $m \to 1$, $R \to 0.287$, $T_1 \to 293$

$V_1 = 0.82442\ m^3$

Process ① $\Rightarrow$ $P_1 V_1^{1.3} = P_2 V_2^{1.3}$, $V_2 = 0.22555\ m^3$

St ② $\Rightarrow$ $P_2 V_2 = m R T_2$

$P_2 \to 5.5 \times 10^2$, $V_2 \to 0.22555$, $R \to 0.287$

$T_2 = 432.25\ K$

$W = \dfrac{P_1 V_1 - P_2 V_2}{n-1} = \boxed{-133.2}\ kJ$ | $Q = W\left(\dfrac{\gamma - n}{\gamma - 1}\right) = -33.3\ kJ$

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$\boxed{10}$ $M = 28\ kg/kmol$, $C_v = 0.741\ kJ/kg\,K$

$$R = \frac{R_o}{M} = \frac{8.3145}{28} = 0.29694\ \frac{kJ}{kg\,K}$$

$$C_v = \frac{R}{\gamma - 1} \quad , \quad 0.741 = \frac{0.29694}{\gamma - 1}$$

$$\boxed{\gamma = 1.4}$$

$$C_p = \gamma C_v = 0.75221\ kJ/kg\,K$$

st ①

$P_1 = 35 \times 10^2\ kPa$

$T_1 = 200 + 273 = 473\ K$

$V_1 = 0.03\ m^3$

adiabatic

$PV^{\gamma} = C$

st ②

$V_2 = 0.09\ m^3$

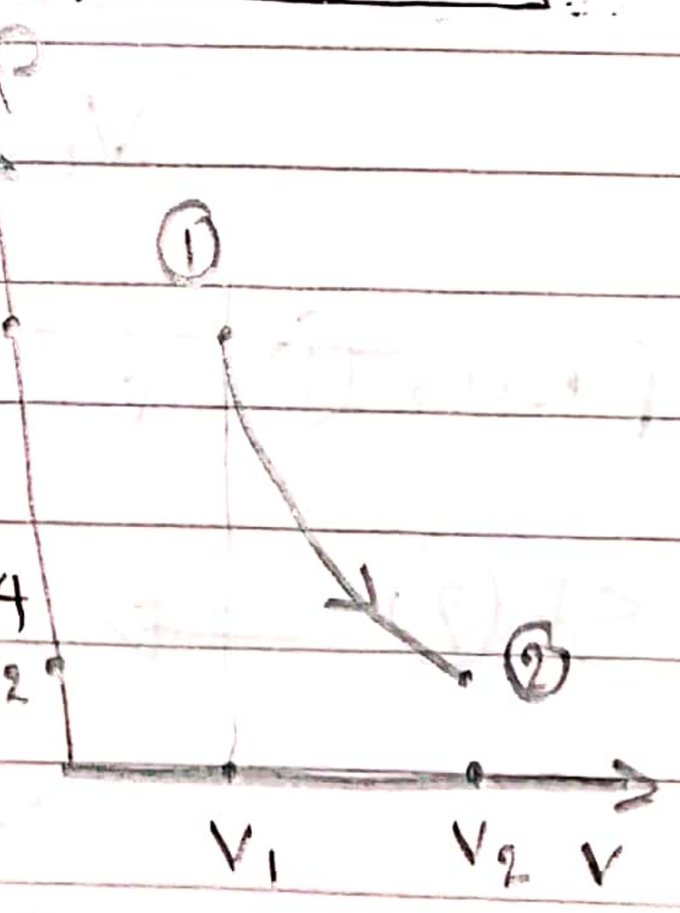
st ① $\Rightarrow$ $P_1 V_1 = m R T_1$

$M = 0.07475\ kg$

Process ① $\Rightarrow$ $P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$

$\gamma \rightarrow 1.4$, $\gamma \rightarrow 1.4$

$P_1 \rightarrow 35 \times 10^2$, $V_1 \rightarrow 0.03$, $V_2 \rightarrow 0.09$



$P_2 = 75.179\ kPa$

st ② $\Rightarrow$ $P_2 V_2 = m R T_2$

$T_2 = 304.8\ K$

$$W = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = 9.334\ kJ$$

$\boxed{8}$ $m = 1\,kg$, Air $\begin{cases} \gamma = 1.4 \\ c_p = 1.005\ \frac{kJ}{kg\,K} \\ C_v = 0.718\ \frac{kJ}{kg\,K} \\ R = 0.287\ \frac{kJ}{kg\,K} \end{cases}$

St ①	isothermally	St ②
$P_1 = 1 \times 10^2\ kP$ $T_1 = 30 + 273 = 303\ K$	$T = C$ $PV = C$ $\longrightarrow$ $T_1 = T_2$	$P_2 = 5 \times 10^2\ kP$ $T_2 = 303\ K$

St ① $\Longrightarrow$ $\underbrace{P_1}_{10^2} V_1 = \underbrace{m}_{1} \underbrace{R}_{0.287} \underbrace{T_1}_{303}$

$V_1 = 0.86961\ m^3$

Process ① $\Longrightarrow$ & $T = C$, $\therefore T_1 = T_2 = 303\ K$

& $PV = C$

$\therefore P_1V_1 = P_2V_2$, $\therefore V_2 = 0.17392\ m^3$

$$W = Q = P_1V_1 \ln \frac{P_1}{P_2} = -140\ kJ$$